

MC-Core Logic Inventory

One channel. Status: **DONE** / RAW (unoptimized) / **STUB** / **OPEN**. Companion to the stage plan + figs fig_00..52.

Block	What it does (one line)	Status
1. Ingress (per rank lane, ×N_RANKS)		
R/W_REQ_AFIFO	Async request in from CIF – the only true CDC in.	done
RD/WR_BP_FIFO	Per-rank back-pressure FIFOs, split by direction (depth 16).	done
WLR_SEL + TURN_CTRL	Weighted R/W admission: batch_dir + batch/age counter; drives the single combined path.	done
straggler adaptive-batch	Shorten new batch when old-dir stragglers pending; sweep on flip-back (bus-level dir only).	judged
LOOKAHEAD	4-entry bypass window (e0..e3 + bank_cmp + prio_enc); younger-to-ready-bank bypass.	done
BANK_DEC	1-of-32 decode to {rank,bank}.	done
2. Per-bank + arb tree		
BANK cell / FSM_TABLE	Per-{rank,bank} state: state[4], open_row, pre_pending, valid, ref_pending, last_op.	done
CMD_ARB	Per bank: pick cas/act/pre by weights (CAS_W row-lock, ACT_W age, PRE_W close-row) + W_MAX.	done
BGB / BG / RANK arb	bank→BG→rank selection; hit-bias + diff-BG rotation + tRTRS-weighted L3.	done
class-split + L3	cas/act/pre lanes, 2:1-per-class across ranks; N-winner bundle (max 3, +ref).	done
p_bit (ref drain)	ref_pending as a priority MSB into CAS arb so target banks drain first.	done
weights: age/cap/ref	4-bit age-of-need, saturates at cap; across ref: no reset/freeze, cap fences both ways.	done
3. Scoreboard / timestamp path		
GC	CK-valued timebase, GC += gear per mc_clk; the only gear-aware element.	decided
scope tables	BANK / BGS (same-BG) / BGD (cross-BG) / RANK / GLOBAL-DQ next_* stores.	done
writeback (_UPD)	On issue stamp next_* = (GC+phase_off)+const per fig_50; ACT bank-level drawn.	RAW
CAS/PRE/other-scope comparators	Extend the same UPD/compare to CAS, PRE, and BGS/BGD/RANK writes.	PENDING
TIMING_CONST	can_x = (GC-next_x)[MSB]==0 (≥, wrap-safe); never ==/XOR.	decided
	JEDEC addends (tRC/tRCD/...) – one shared copy, feeds every UPD adder.	done
4. Phaser / packer		
PHASE_PACKER	Token domain: 1 token/command {cmd,bank,addr,phase_off,len}; assigns offsets.	done
CA-beat encoder	Expands each token to 1–2 dfi phases; tail_pending straddles 2 bundles at 1:1.	done
gear mask + gear reg	Slot mask by gear (fill 0..gear-1, rest NOP); gear = preset config reg.	done
DFI_MUX	init FSM vs scheduler encoder, sel = init_done (one-way). Only 2 DFI drivers.	done
5. Data path (write + read)		
WR_ACCUM	Burst accumulator: AXI beats → 64B line + WSTRB; narrow flag picks DM-mask vs full.	done
WD_SRAM + wr_ptr	Allocate-on-arrival write buffer, incremental order; back-pressure on full.	done
WRITE_ROB tracker	Per aw_id {vld, N, gen_idx, done_cnt}: gen_idx→addr offset, retire at done_cnt==N.	done
WR_SPLIT + WL_LAUNCH	512b → 2×gear beats to dfi_wrdata (CWL-delayed).	done
RD_CAP + RD_ACCUM	dfi_rddata beats → 64B → one 512b write to RD_SRAM (RL line).	done
POP_CTRL	Free the SRAM line only past the ALERT_n window (write-CRC retry from SRAM).	done
COMPLETION	Merge rd_done/wr_done → resp AFIFO, tag-only {rob_index,pkt,status}.	done
ECC / RMW	On-die ECC → DM mask (no RMW); controller ECC → RMW read-allocate + interlock.	open
Segmentation owner	CIF pre-segment vs MC-side WR_ACCUM packetization – pick one.	open
6. Maintenance engine (peer, never CAS; flags to scheduler)		
INIT FSM (16)	DDR5 power-on: reset→MRW-train→ZQCL→training→init_done (drives DFI till done).	done
REFRESH FSM (6)	Leaky-bucket credits; REFsb-preferred, drain (policy B) → PRE → REF; gate_rfc.	done
RFM FSM (6)	Per-bank RAA (+ACT, –ref); RAA≥RAAIMT → RFMsb/ab.	done
ZQCAL FSM (7)	MPC ZQ start/latch; gate_zq (rank).	done
MR_POLL FSM (6)	Periodic MRR(MR4); TUF=1 (>85C) → tREFI/2 (temp reload).	done
MR_WRITE FSM (9)	CSR mode-reg write + optional verify + timing-apply; shares MR Read Arbiter.	done
POWER_MGMT (10)	Power-down + self-refresh branches (CKE, T_XP/T_XS/T_DLLK).	done

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MR Read Arbiter	Grants 1 of {MR_POLL, MR_WRITE} the MRR sideband; mrr_requester tags return.	done
FREQ_CHANGE FSM	DVFS: quiesce→self-refresh→DFI freq-change handshake→PHY relock→reload timing_reg_file + MRW + ZQ→SRX→update gear.	stub
ERROR / ALERT_n	Watch dfi_alert_n (write-CRC/CA-parity); single recovery point.	stub
7. Still open / to write		
Address Map Unit	daddr → {rank,bg,bank,row,col}; is any hash left MC-side (vs CIF)?	open
RAW-hazard doc	Same-line R/W ordering owned by CIF (MC = pure reorderable engine); write the doc.	PENDING
Packer detail	Bundle-assembly rules, ≥2-live-BG preference, spill handling.	PENDING
fig_52 comparator	Redraw as (GC+phase)≥next + “GC=CK, += gear” note.	minor